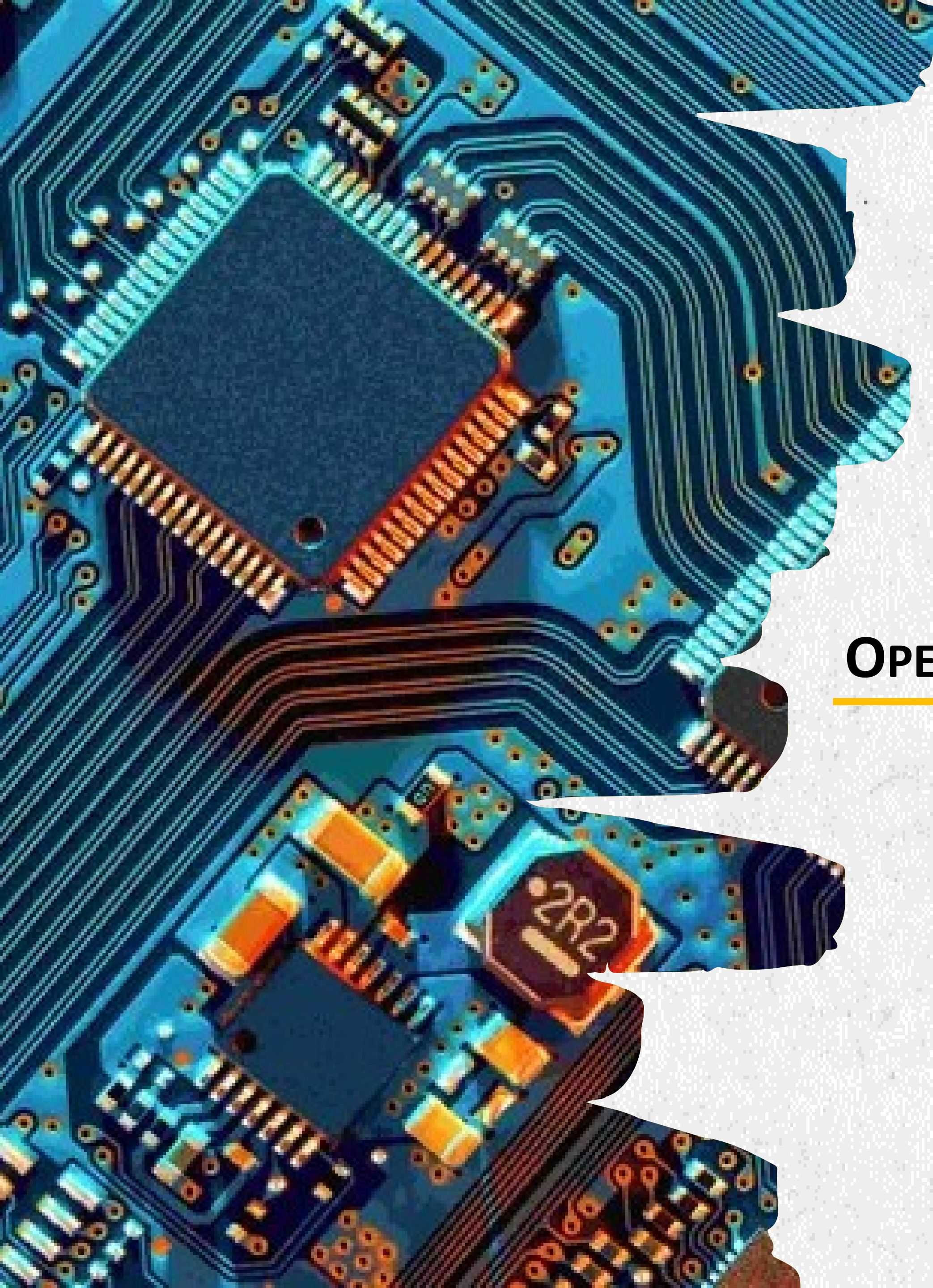


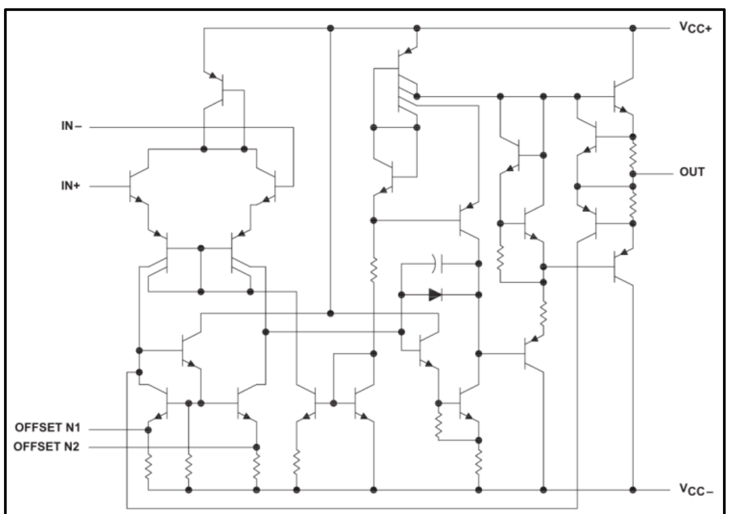
CHAPTER 9

OPERATIONAL AMPLIFIER (OP-AMP)

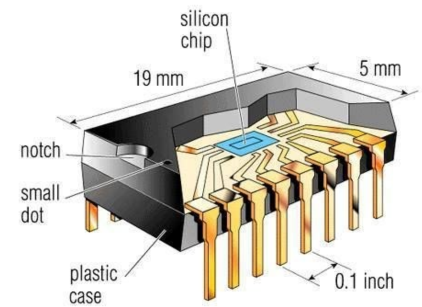
INTRODUCTION

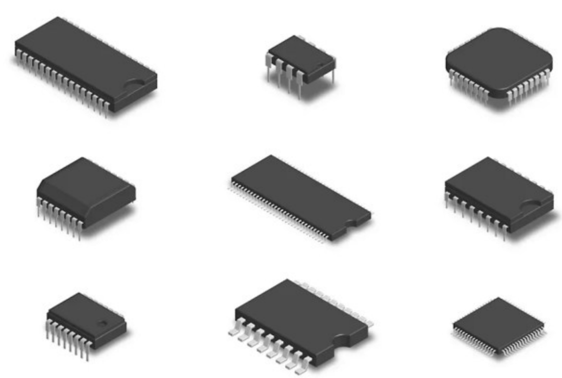
- In the previous chapters, you have studied *discrete components 離散元件, i.e., resistors, capacitors, inductors, diodes, and transistors.* They must be *interconnected* in a circuit with other devices to form ***a complete functional unit***.
- Now, we will begin the study of *integrated circuits (ICs、積體電路)*, where many resistors, capacitors, diodes, transistors are *embedded on a single tiny chip 晶片*.

Equivalent circuit of Op-Amp
(運算放大器的等效電路)



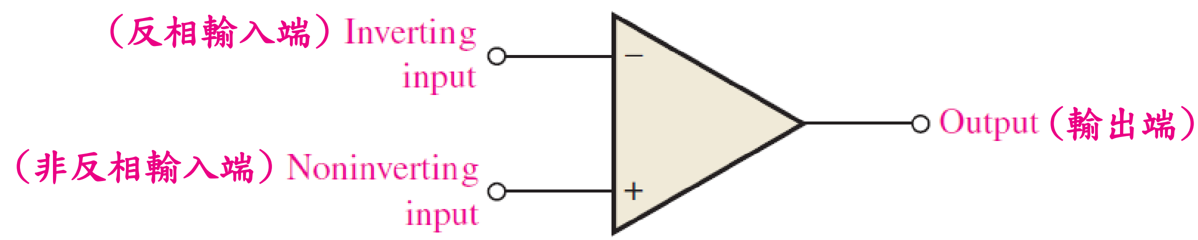
Component Count	
Transistors	22
Resistors	11
Diode	1
Capacitor	1

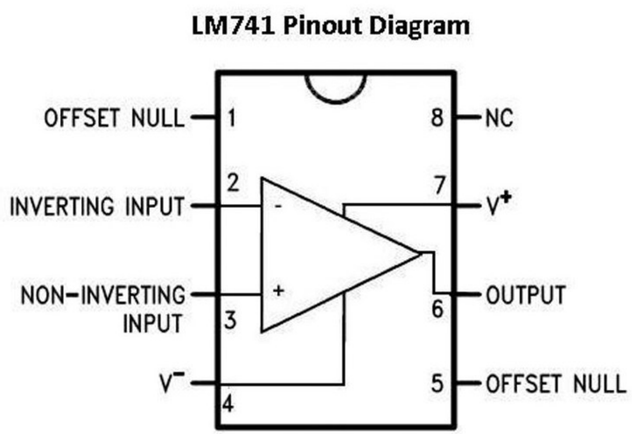


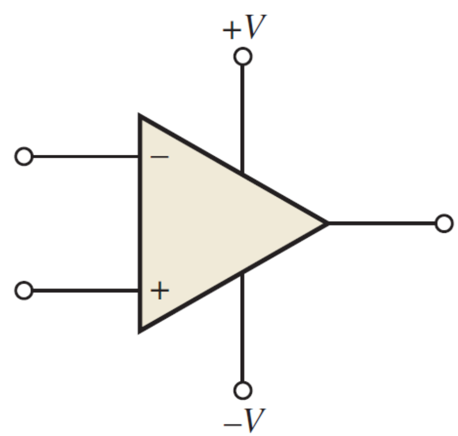


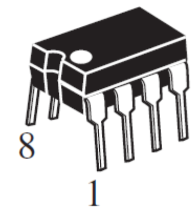
OPERATIONAL AMPLIFIER (OP-AMP)

- *Operational Amplifiers (Op-Amp、運算放大器)* are one of the most popular ICs. They have been used *to perform mathematical operations 數學運算*, such as addition (+), subtraction (-), integration (∫), and differentiation (d/dt) – thus the term *"operational 運算"*.





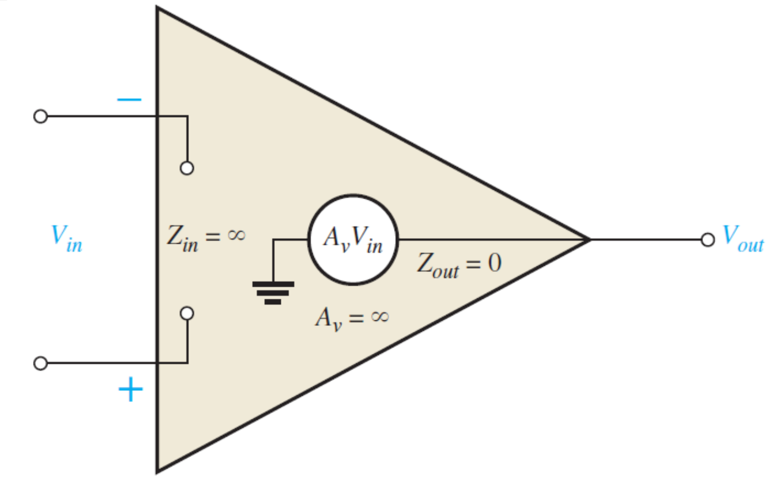




Dual-in line package (DIP、雙列直插封裝)

Surface-mount technology (SMT、表面貼裝技術)

OP-AMP CHARACTERISTICS

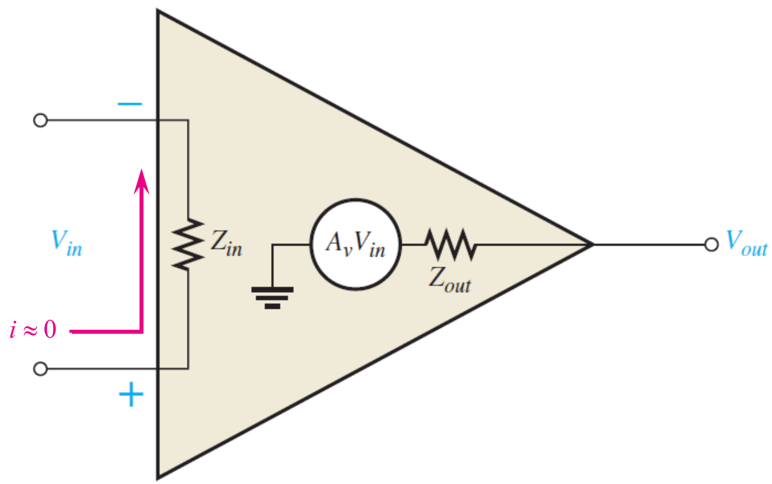


Ideal model

- Infinite input impedance ($Z_{in} = \infty$)
- Zero output impedance ($Z_{out} = 0$)
- Infinite voltage gain ($A_v = \infty$)

$$V_{out} = A_v(V_{in}^{+} - V_{in}^{-})$$

"We will use the ideal model in our study"



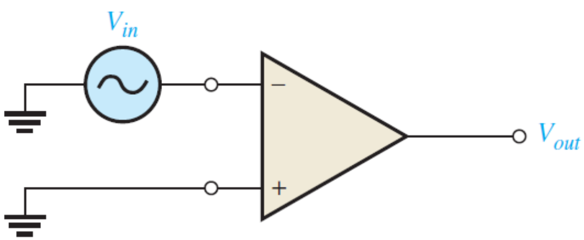
Practical model

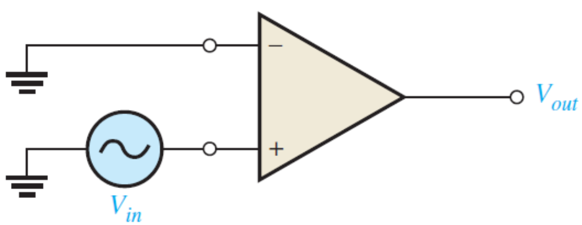
- Very high input impedance ($Z_{in} \to \infty$)
- Very low output impedance ($Z_{out} \approx 0$)
- Very high voltage gain ($A_v \to \infty$)
  [A_v is typically between 10,000 ~ 500,000]

$$V_{out} \le A_v(V_{in}^{+} - V_{in}^{-})$$

OP-AMP INPUT MODES

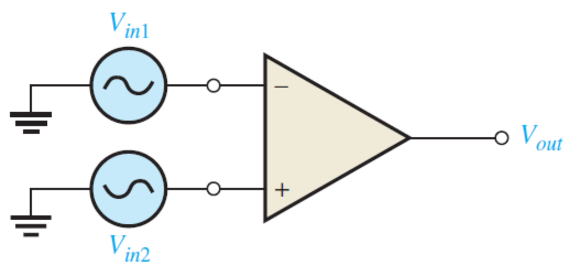
Single-ended differential mode (單端差模)

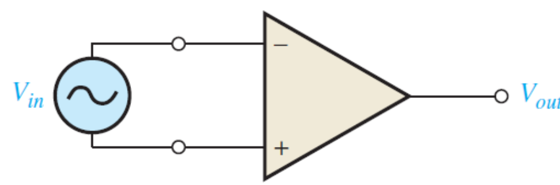




"Amplify the inverted or non-inverted input signal"

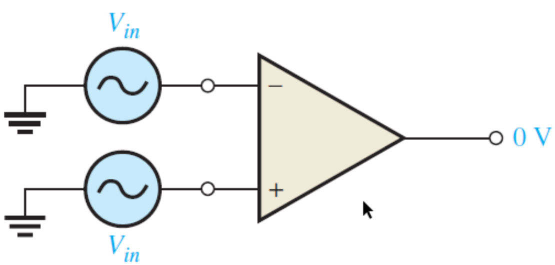
Double-ended differential mode (雙端差模)





"Amplify the difference between the two inputs"

Common mode (共模)

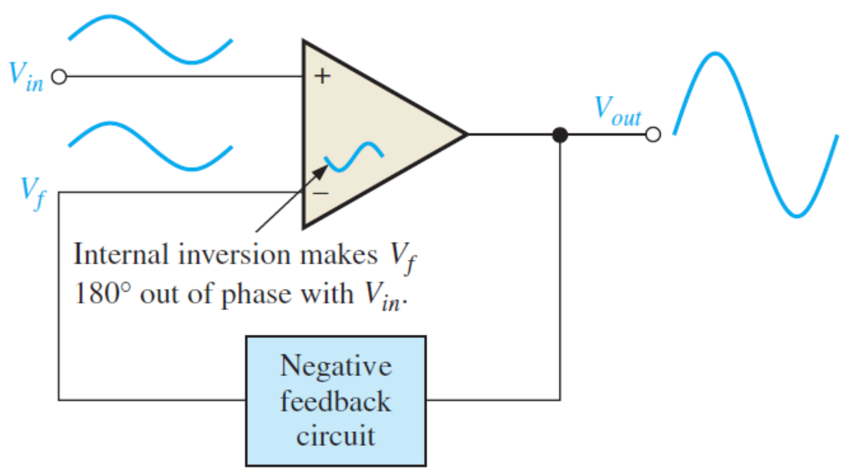


"Cancel out the unwanted signal (noise)"

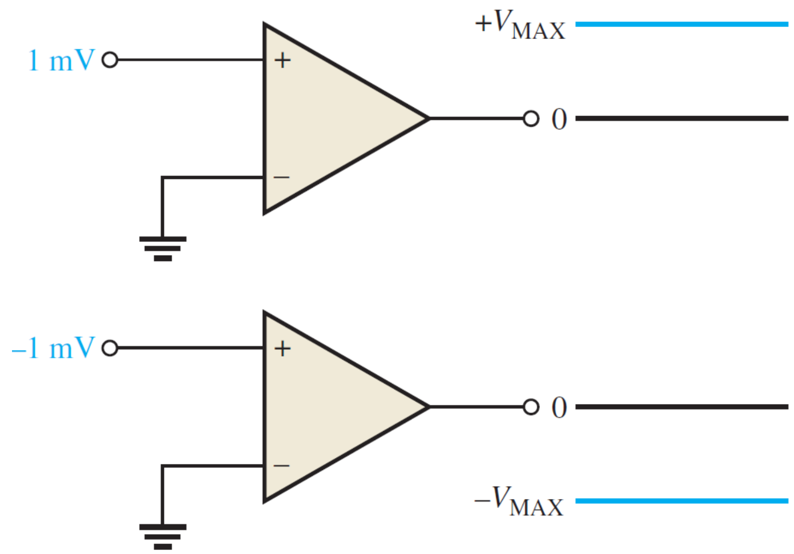
OPEN-LOOP AND CLOSED-LOOP

- *Op-Amps* have *many operational circuits* (運算電路), most of which are *negative feedback* (負回饋) connections.
- If the *Op-Amps* operate *with open-loop connection*, the application is limited to *only comparator* (比較器) *applications*.

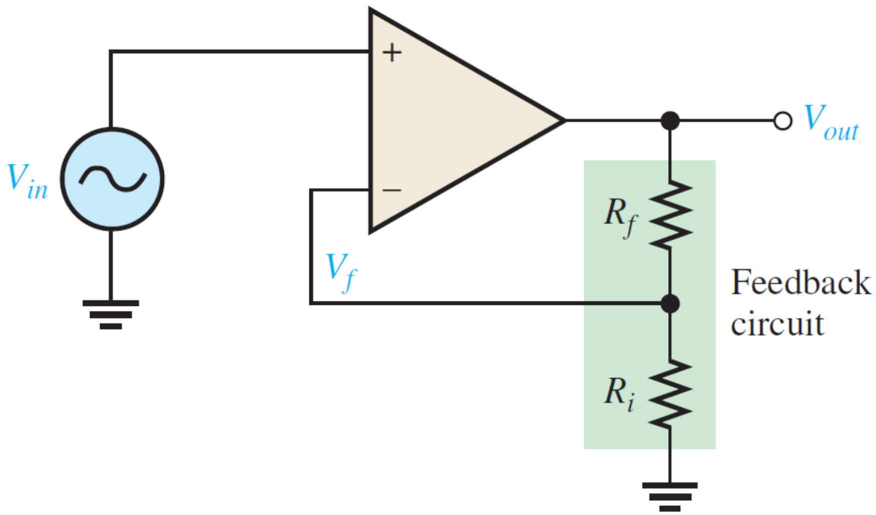
Negative feedback
(Closed-loop 閉迴路)



Open-loop (開回路)



NONINVERTING AMPLIFIER 非反相放大器



Open-loop gain (A_{ol})

$$V_{out} = A_{ol}(V_{in} - V_f)$$

Open-loop gain (A_{cl})

$$V_{out} = A_{cl}V_{in}$$

A_{ol} is known from the datasheet.
How can we know A_{cl}?

- Step 1:

$$V_{out} = A_{ol}(V_{in} - V_f) = A_{ol}V_{in} - A_{ol}BV_{out}$$

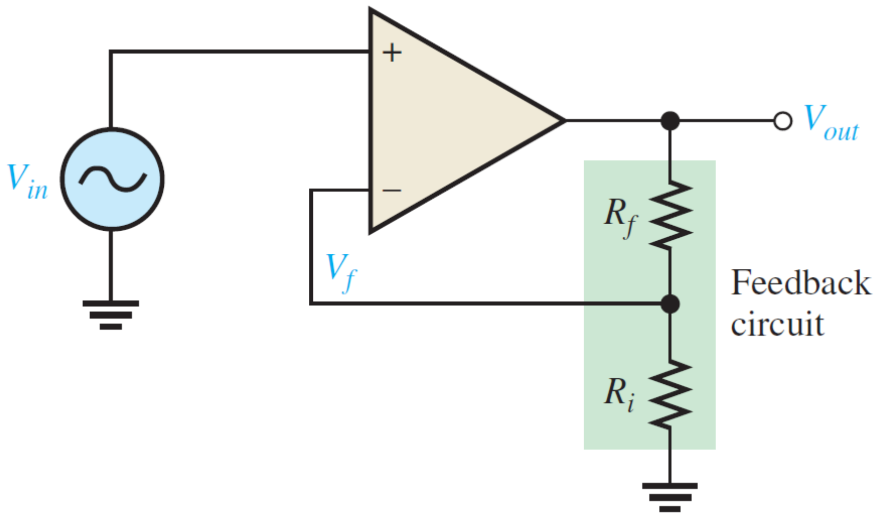
$$V_f = \underbrace{\frac{R_i}{R_i + R_f}}_{B} V_{out}$$

- Step 2:

$$\frac{V_{out}}{V_{in}} = \lim_{A_{ol}\to\infty} \frac{A_{ol}}{1 + A_{ol}B} \cong \frac{A_{ol}}{A_{ol}B} = \frac{1}{B}$$

$$\frac{V_{out}}{V_{in}} = \frac{R_i + R_f}{R_i} = \left. 1 + \frac{R_f}{R_i} \right\} A_{cl}$$

NONINVERTING AMPLIFIER 非反相放大器



$$\frac{V_{out}}{V_{in}} = \frac{R_i + R_f}{R_i} = \underbrace{1 + \frac{R_f}{R_i}}_{A_{cl}}$$

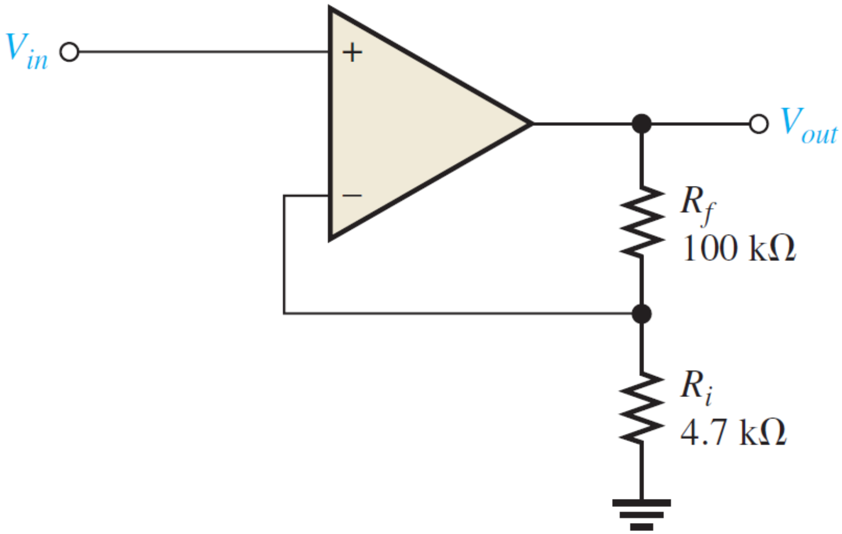
$$V_f = \underbrace{\frac{R_i}{R_i + R_f}}_{B} V_{out} \quad \Rightarrow \quad V_f = V_{in}$$

(Only for negative feedback)

- Simple derivation:

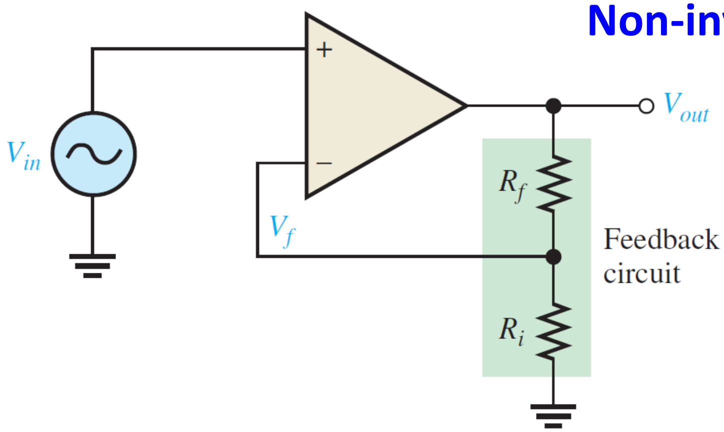
$$V_{in} = \frac{R_i}{R_i + R_f} V_{out} \quad \rightarrow \quad V_{out} = \frac{R_i + R_f}{R_i} V_{in} = \underbrace{\left(1 + \frac{R_f}{R_i}\right)}_{A_{cl}} V_{in} \quad \rightarrow \quad A_{cl} = 1 + \frac{R_f}{R_i}$$

EXERCISE



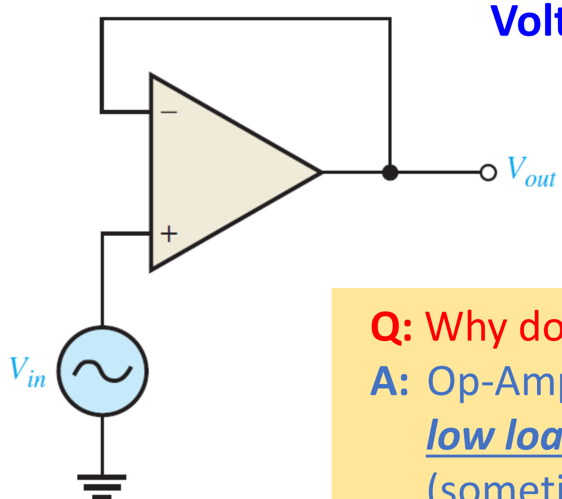
1. Find the closed-loop gain A_{cl}?
2. If $V_{in} = 0.5\ \mathrm{V}$, what is V_{out}?

VOLTAGE-FOLLOWER 電壓隨耦器



Non-inverting amplifier (非反相放大器)

$$\underbrace{\frac{V_{out}}{V_{in}} = 1 + \frac{R_f}{R_i}}_{A_{cl}}$$



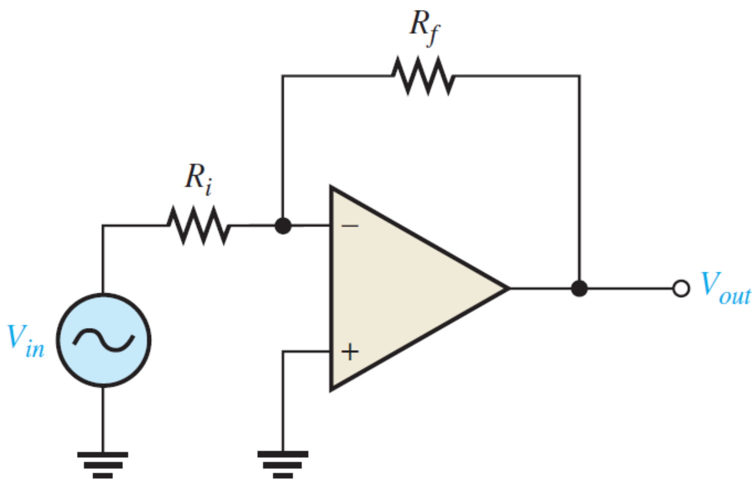
Voltage-follower (電壓隨耦器)

$$A_{cl} = \frac{V_{out}}{V_{in}} = 1$$

Q: Why do we need the ***voltage-follower***?

A: Op-Amp has ***high input impedance* 輸入阻抗**, leading to ***low load* 負載 *during signal measurement***.
(sometimes called a ***buffer* 緩衝放大器**)

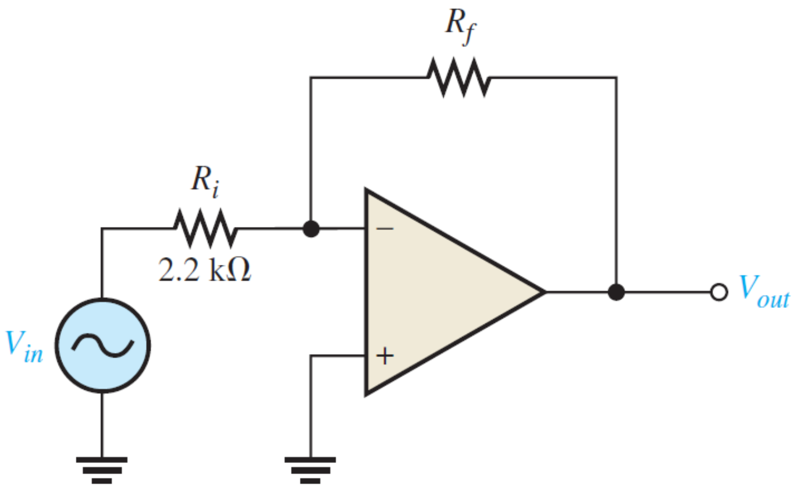
INVERTING AMPLIFIER 反相放大器



- Find the closed-loop gain A_{cl}

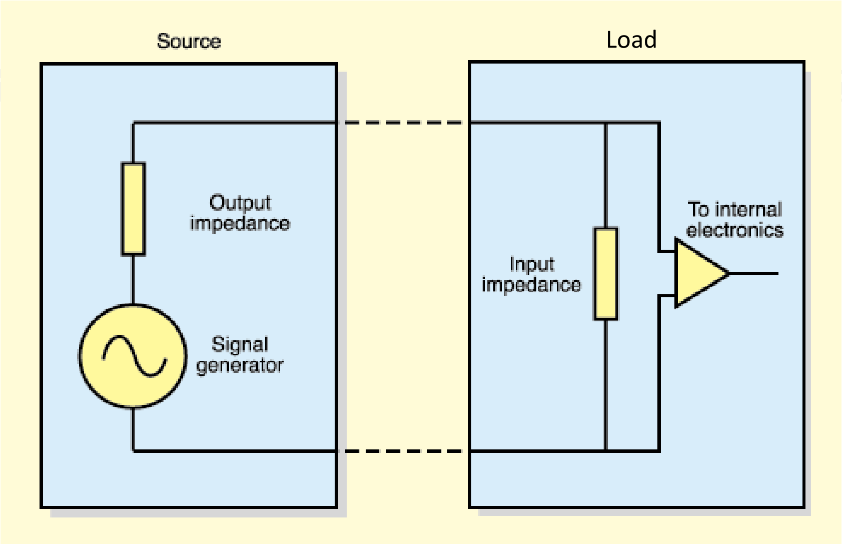
$$A_{cl} = \frac{V_{out}}{V_{in}} = -\frac{R_f}{R_i}$$

EXERCISE



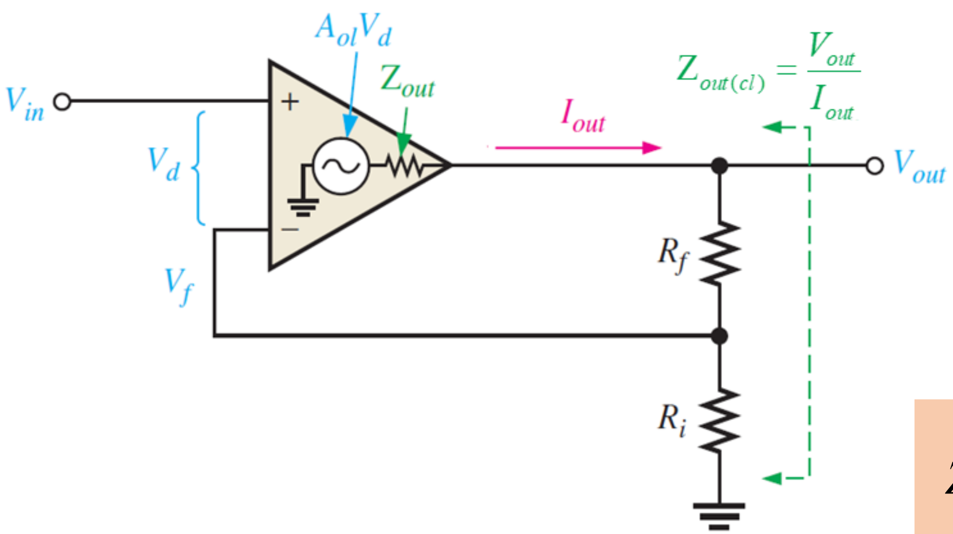
- Determine the value of R_f such that the gain $A_{cl} = -100$.

INPUT AND OUTPUT IMPEDANCE 輸入出阻抗



- Input impedance 輸入阻抗 – the impedance seen by the source, i.e., *how much current the input will draw from the source*.
  - High input imp. (or low load) – the load draws very little current, not affecting the source.
  - Low input imp. (or high load) – the load draw high current, causing the voltage drop at the source.
- Output impedance 輸出阻抗 – the impedance seen by the load, i.e., *how much the source resists to supply current*.
  - High output imp. – the output voltage drops when a load is connected.
  - Low output imp. – the output voltage remains the same, despite heavy load.

I/O IMPEDANCE OF NONINVERTING AMPLIFIER



Input impedance analysis

$$B = \frac{R_i}{R_f + R_i}$$

$$V_{in} = V_d + V_f$$

$$V_{in} = V_d + BV_{out} = V_d + A_{ol}BV_d = (1 + A_{ol}B)V_d$$

$$V_{in} = (1 + A_{ol}B)I_{in}Z_{in}$$

$$Z_{in(cl)} = \frac{V_{in}}{I_{in}} = (1 + A_{ol}B)Z_{in}$$

Output impedance analysis

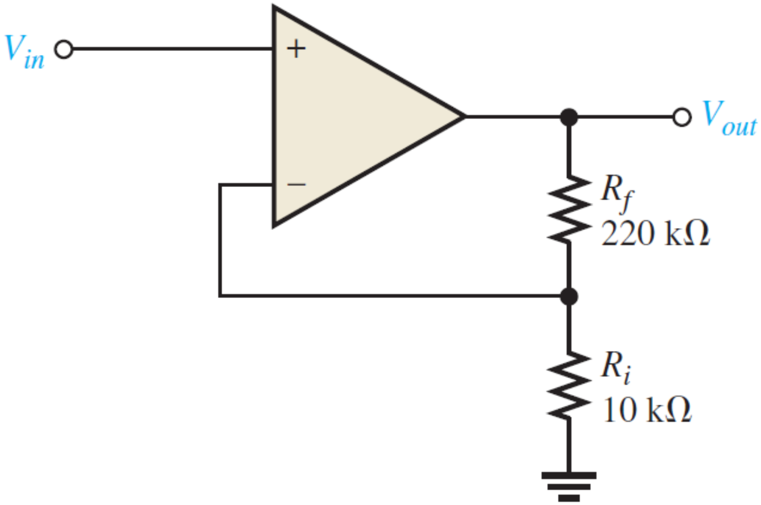
$$V_{out} = A_{ol}V_d - I_{out}Z_{out} = A_{ol}(V_{in} - BV_{out}) - I_{out}Z_{out}$$

$$V_{out} = \frac{A_{ol}V_{in} - I_{out}Z_{out}}{1 + A_{ol}B} \rightarrow Z_{out(cl)} = \left|\frac{dV_{out}}{dI_{out}}\right| = \frac{Z_{out}}{1 + A_{ol}B}$$

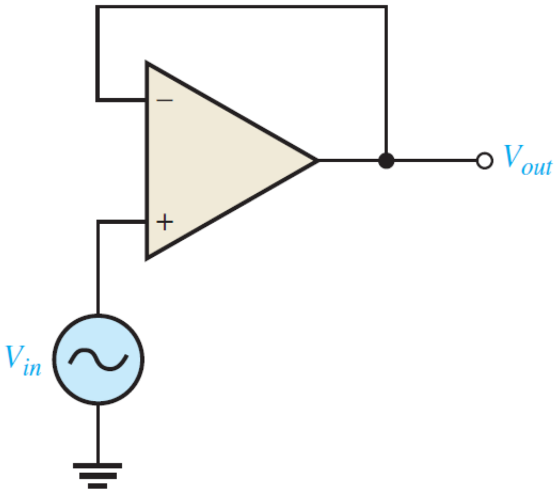
EXERCISE

- The Op-Amp datasheet gives $Z_{in} = 2\ \text{M}\Omega$, $Z_{out} = 75\ \Omega$, and $A_{ol} = 200{,}000$.

1. Determine $Z_{in(cl)}$ and $Z_{out(cl)}$.
2. Find the closed-loop gain A_{cl}



I/O IMPEDANCE OF VOLTAGE-FOLLOWER



Input impedance analysis

$$Z_{in(cl)} = \frac{V_{in}}{I_{in}} = (1 + A_{ol}B)Z_{in}$$

$$B = \frac{R_i}{R_f + R_i} = 1$$

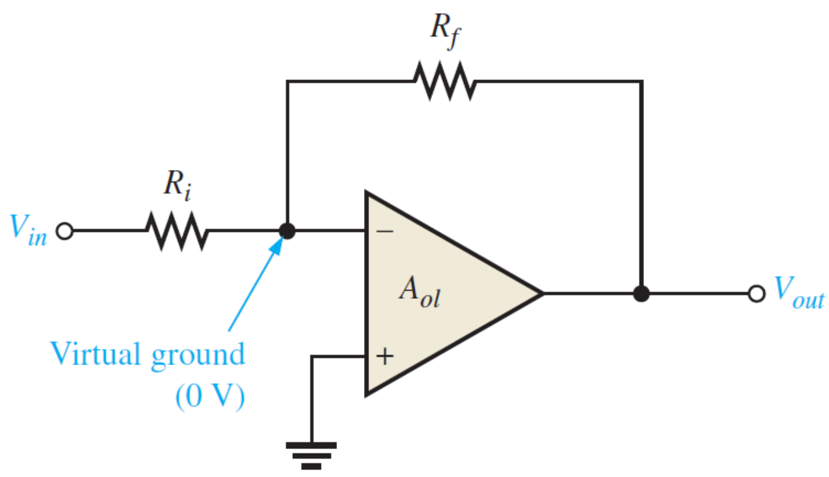
$$Z_{in(cl)} = (1 + A_{ol})Z_{in}$$

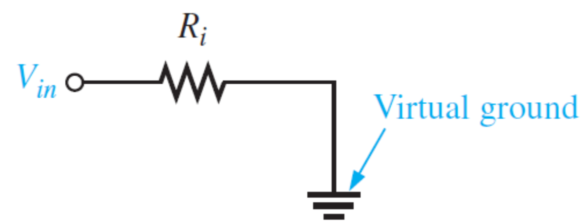
Output impedance analysis

$$Z_{out(cl)} = \frac{Z_{out}}{1 + A_{ol}B}$$

$$Z_{out(cl)} = \frac{Z_{out}}{1 + A_{ol}}$$

I/O IMPEDANCE OF INVERTING AMPLIFIER





Input impedance analysis

$$Z_{in(cl)} \cong R_i$$

Output impedance analysis

$$V_{out} = A_{ol}V_d - I_{out}Z_{out} \quad = -A_{ol}V_f - I_{out}Z_{out}$$

$$V_{out} = -A_{ol}[BV_{out} + (1-B)V_{in}] - I_{out}Z_{out}$$

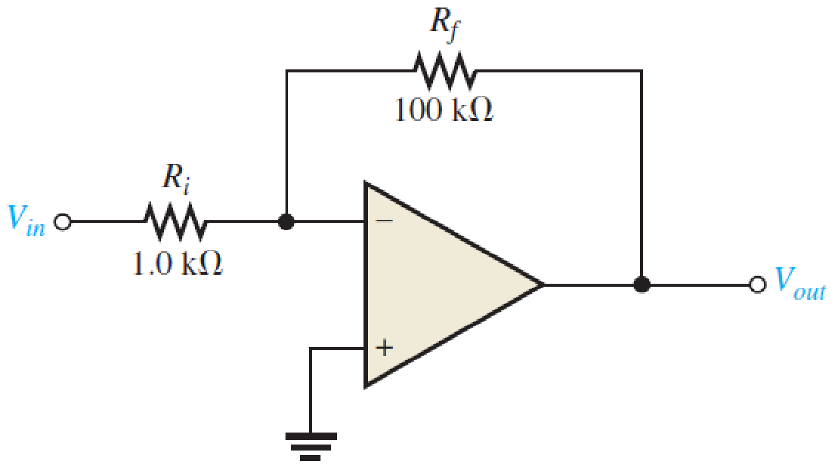
$$V_{out} = \frac{-A_{ol}(1-B)V_{in} - I_{out}Z_{out}}{1 + A_{ol}B} \rightarrow \boxed{Z_{out(cl)} = \left|\frac{dV_{out}}{dI_{out}}\right| = \frac{Z_{out}}{1 + A_{ol}B}}$$

(same as the noninverting amplifier)

EXERCISE

- The Op-Amp datasheet gives $Z_{in} = 4\ \mathrm{M\Omega}$, $Z_{out} = 50\ \Omega$, and $A_{ol} = 50{,}000$.

1. Determine $Z_{in(cl)}$ and $Z_{out(cl)}$.
2. Find the closed-loop gain A_{cl}



μA741 General-Purpose Operational Amplifiers

1 Features

- Short-Circuit Protection
- Offset-Voltage Null Capability
- Large Common-Mode and Differential Voltage Ranges
- No Frequency Compensation Required
- No Latch-Up

2 Applications

- DVD Recorders and Players
- Pro Audio Mixers

3 Description

The μA741 device is a general-purpose operational amplifier featuring offset-voltage null capability.

The high common-mode input voltage range and the absence of latch-up make the amplifier ideal for voltage-follower applications. The device is short-circuit protected and the internal frequency compensation ensures stability without external components. A low-value potentiometer may be connected between the offset null inputs to null out the offset voltage as shown in Figure 12.

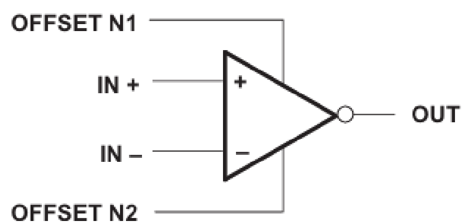
The μA741C device is characterized for operation from 0°C to 70°C.

Device Information(1)

PART NUMBER	PACKAGE	BODY SIZE (NOM)
μA741CD	SOIC (8)	4.90 mm × 3.91 mm
μA741CP	PDIP (8)	9.81 mm × 6.35 mm
μA741CPS	SO (8)	6.20 mm × 5.30 mm

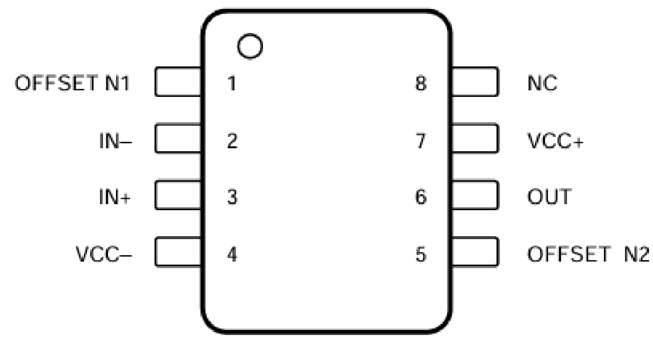
(1) For all available packages, see the orderable addendum at the end of the data sheet.

Simplified Schematic



5 Pin Configurations and Functions

uA741C D, P, or PS Package
8-Pin SOIC, PDIP, SO
Top View



Not to scale

NC- no internal connection

Pin Functions

PIN		I/O	DESCRIPTION
NAME	NO.		
IN+	3	I	Noninverting input
IN–	2	I	Inverting input
NC	8	—	No internal connection
OFFSET N1	1	I	External input offset voltage adjustment
OFFSET N2	5	I	External input offset voltage adjustment
OUT	6	O	Output
VCC+	7	—	Positive supply
VCC–	4	—	Negative supply

6.1 Absolute Maximum Ratings

over virtual junction temperature range (unless otherwise noted)[1]

			MIN	MAX	UNIT
Supply voltage, V_{CC}[2]		μA741C	–18	18	V
Differential input voltage, V_{ID}[3]		μA741C	–15	15	V
Input voltage, V_I (any input) [2][4]		μA741C	–15	15	V
Voltage between offset null (either OFFSET N1 or OFFSET N2) and V_{CC-}		μA741C	–15	15	V
Duration of output short circuit[5]			Unlimited		
Continuous total power dissipation			See *Thermal Information*		
Case temperature for 60 seconds		μA741C	N/A	N/A	°C
Lead temperature 1.6 mm (1/16 inch) from case for 60 seconds		μA741C	N/A	N/A	°C
Lead temperature 1.6 mm (1/16 inch) from case for 10 seconds	D, P, or PS package	μA741C		260	°C
Operating junction temperature, T_J				150	°C
Storage temperature range, T_{stg}		μA741C	–65	150	°C

6.2 Recommended Operating Conditions

			MIN	MAX	UNIT
V_{CC+}	Supply voltage		5	15	V
V_{CC-}			–5	–15	
T_A	Operating free-air temperature	μA741C	0	70	°C

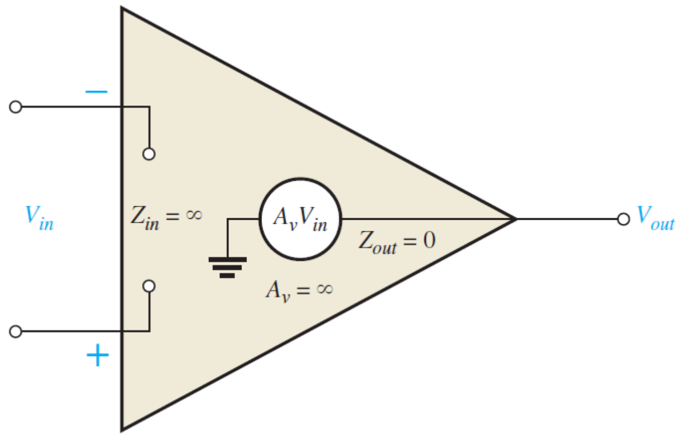
6.4 Electrical Characteristics: μA741C

at specified virtual junction temperature, $V_{CC\pm}$ = ±15 V (unless otherwise noted)

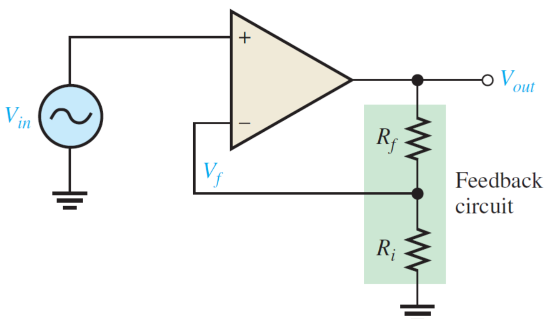
PARAMETER		TEST CONDITIONS[1]		MIN	TYP	MAX	UNIT
V_{IO}	Input offset voltage	V_O = 0	25°C		1	6	mV
			Full range			7.5	
$\Delta V_{IO(adj)}$	Offset voltage adjust range	V_O = 0	25°C		±15		mV
I_{IO}	Input offset current	V_O = 0	25°C		20	200	nA
			Full range			300	
I_{IB}	Input bias current	V_O = 0	25°C		80	500	nA
			Full range			800	
V_{ICR}	Common-mode input voltage range	25°C		±12	±13		V
		Full range		±12			
V_{OM}	Maximum peak output voltage swing	R_L = 10 kΩ	25°C	±12	±14		V
		R_L ≥ 10 kΩ	Full range	±12			
		R_L = 2 kΩ	25°C	±10			
		R_L ≥ 2 kΩ	Full range	±10			
A_{VD}	Large-signal differential voltage amplification	R_L ≥ 2 kΩ	25°C	20	200		V/mV
		V_O = ±10 V	Full range	15			
r_i	Input resistance	25°C		0.3	2		MΩ
r_o	Output resistance	V_O = 0; see[2]	25°C		75		Ω
C_i	Input capacitance	25°C			1.4		pF
CMRR	Common-mode rejection ratio	$V_{IC} = V_{ICRmin}$	25°C	70	90		dB
			Full range	70			
k_{SVS}	Supply voltage sensitivity ($\Delta V_{IO}/\Delta V_{CC}$)	V_{CC} = ±9 V to ±15 V	25°C		30	150	μV/V
			Full range			150	
I_{OS}	Short-circuit output current	25°C			±25	±40	mA
I_{CC}	Supply current	V_O = 0; no load	25°C		1.7	2.8	mA
			Full range			3.3	
P_D	Total power dissipation	V_O = 0; no load	25°C		50	85	mW
			Full range			100	

SUMMARY

- Op-Amp have
  1. High input impedance
  2. Low output impedance
  3. High voltage gain



Non-inverting amplifier

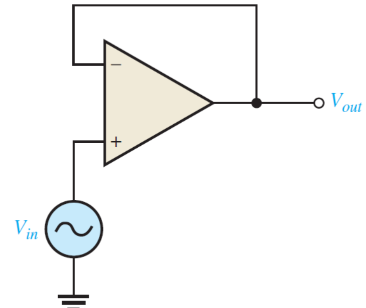


$$A_{cl} = 1 + \frac{R_f}{R_i}$$

$$Z_{in(cl)} = (1 + A_{ol}B)Z_{in}$$

$$Z_{out(cl)} = \frac{Z_{out}}{1 + A_{ol}B}$$

Voltage-follower

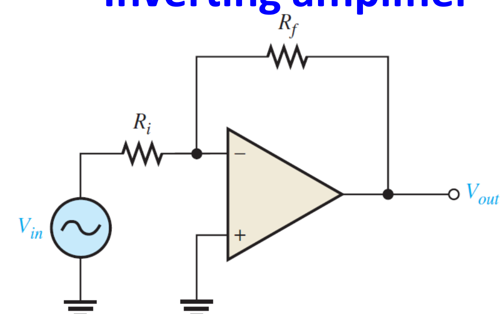


$$A_{cl} = \frac{V_{out}}{V_{in}} = 1$$

$$Z_{in(cl)} = (1 + A_{ol})Z_{in}$$

$$Z_{out(cl)} = \frac{Z_{out}}{1 + A_{ol}}$$

Inverting amplifier



$$A_{cl} = -\frac{R_f}{R_i}$$

$$Z_{in(cl)} \cong R_i$$

$$Z_{out(cl)} = \frac{Z_{out}}{1 + A_{ol}B}$$